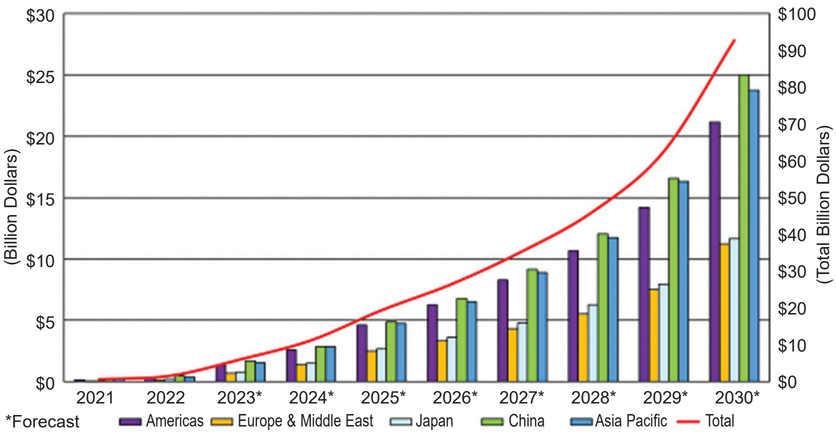
## RISC-V SoC regional forecasts

Here are the regional market share forecasts for RISC-V SoCs:

- **Americas:** Revenue of $1.3 billion in 2023, with market share increasing slightly from 2.6% to 2.7% by 2030
- **Europe & EMEA:** Market share growing from 7.4% in 2023 to 9.9% by 2030
- **Japan:** Market share expanding from 7.1% in 2023 to 18% by 2030
- **China:** Market share starting at 33.8% in 2023 and only slightly increasing to 34% by 2030
- **Asia Pacific:** Market share decreasing from 45.5% in 2023 to 32.6% by 2030




*Total RISC-V SoC regional revenues 2021-2030 (Source: The SHD Group, January 2024)*

The RISC-V market's application segment shares are projected to change as follows by 2030:

- **Industrial:** Remains relatively stable, from 2.6% in 2023 to 2.7%
- **Automotive:** Increases from 7.4% in 2023 to 9.9%
- **Networking:** Grows significantly from 7.1% in 2023 to 18%
- **Computer:** Slightly declines from 33.8% in 2023 to 34%
- **Consumer:** Decreases from 45.5% in 2023 to 32.6%
- **Other:** Drops from 3.6% in 2023 to 2.6%

existing SoCs and by an established ecosystem.

With advanced driver assistance systems (ADAS) and artificial intelligence being integrated into automobiles, the automotive segment is undergoing a complete transformation.

***Market shipments.*** RISC-V SoC shipments grew to 1.3 billion units in 2023 and are projected to reach 16.2 billion units by 2030, with a CAGR of 44%.

More IP vendors are joining RISC-V design support. RISC-V specifications are being incorporated into non-RISC-V products. This is primarily due to the level of customisation and cost-effectiveness RISC-V provides. As automation is applied to an increasing number of tasks, demand for chips customised for particular tasks is rising.

The consumer segment leads with 0.7 billion units in 2023, and is expected to reach 8.4 billion by 2030. The computer and automotive segments also show significant growth.

***Integration within functions.*** All SoCs have different CPU cores for different functionalities, including deeply embedded systems (finite state machines), microcontroller units (MCUs), co-processors, and host processors. RISC-V-based CPU cores are being integrated into various functions within SoCs for more efficient, customisable, and cost-effective solutions in applications including consumer electronics, networking, computing, and automotive systems.

The consumer segment sees the highest use of RISC-V-based FSMs and MCUs, while co-processors are increasingly used in AI-driven consumer devices. Host processors are key in running software applications across various platforms. The FSM function dominated in revenue in 2023 and is expected to remain the largest functional category by 2030, with other functions gaining ground as higher-complexity SoCs enter the market.

### RISC-V CPU IP market forecast

Royalty revenues from RISC-V-based CPU IPs are expected to surpass its licensing revenues around 2027, with royalties reaching 1.5 times

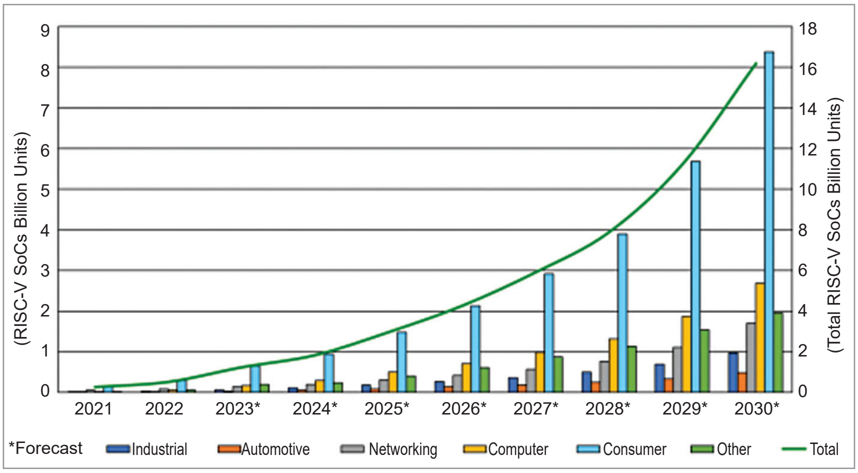



*Fig. 2: Market unit shipments for all RISC-V SoCs by application, 2021-2030 (Source: The SHD Group, January 2024)*